

## NOV-DEC 2025 EXAMINATION

## I B.TECH.

## IT10007: FUNDAMENTALS OF INFORMATION TECHNOLOGY AND ARTIFICIAL INTELLIGENCE

Time 3 Hr.

Max. Marks 70

Note: Each question consists of five options: a, b, c, d, and e. Options a, b, and c are mandatory in every question. From options d and e, attempt either d or e, but not both.

| Question |                                                                                            | Marks | CO | BL | PI    |
|----------|--------------------------------------------------------------------------------------------|-------|----|----|-------|
| Q1       | (a) List any two differences between Data and Information.                                 | 2     | 1  | 2  | 1.4.1 |
|          | (b) Differentiate between Primary and Secondary memory.                                    | 2     | 1  | 2  | 1.4.1 |
|          | (c) Explain what is meant by network topology, giving suitable examples.                   | 3     | 1  | 2  | 1.4.1 |
|          | (d) Answer the following questions.                                                        |       |    |    |       |
|          | a) Write a short note on input, output, and storage devices                                | 3.5   | 1  | 2  | 1.4.1 |
|          | b) What is an operating system? Explain its main functions                                 | 3.5   | 1  | 3  | 1.4.1 |
| OR       |                                                                                            |       |    |    |       |
|          | (e) Explain in detail the block schematic of a digital computer with its working           | 7     | 1  | 3  | 1.4.1 |
| Q2       | (a) Differentiate between keyword and identifier.                                          | 2     | 2  | 2  | 1.4.1 |
|          | (b) Describe arithmetic, relational, and logical operators with examples.                  | 2     | 2  | 3  | 1.4.1 |
|          | (c) Write a C program to swap two numbers without using a temporary variable.              | 3     | 2  | 3  | 1.4.1 |
|          | (d) Fill in the Blanks                                                                     |       |    |    | 1.4.1 |
|          | a) C is a _____ level programming language.                                                | 1     | 2  | 2  |       |
|          | b) The symbol used for assignment is _____.                                                | 1     | 2  | 2  |       |
|          | c) Integer constants do not contain _____ point.                                           | 1     | 2  | 2  |       |
|          | d) ++a is called _____ increment.                                                          | 1     | 2  | 2  |       |
|          | e) In the expression $x = a + b * c$ , the operator evaluated first is _____.              | 1     | 2  | 2  |       |
|          | f) The header file <code>stdio.h</code> is included by writing _____.                      | 1     | 2  | 2  |       |
|          | g) The operator used for modulus operation is _____.                                       | 1     | 2  | 2  |       |
|          | OR                                                                                         |       |    |    |       |
|          | (e) a) Write a C program to find the larger of two numbers using the conditional operator. | 4     | 2  | 6  | 1.4.1 |
|          | b) Write a C program to evaluate the expression $x = a + b * c - d / e$ .                  | 3     | 2  | 4  | 1.4.1 |
| Q3       | (a) Write two differences between break and continue.                                      | 2     | 3  | 2  | 1.4.1 |
|          | (b) What is a pointer? Give its syntax.                                                    | 2     | 3  | 2  | 1.4.1 |
|          | (c) Explain break, continue, and goto with examples.                                       | 3     | 3  | 4  | 1.4.1 |
|          | (d) Fill in the Blanks                                                                     |       |    |    | 1.4.1 |
|          | a) The loop that executes at least once is the _____ loop.                                 | 1     | 3  | 2  |       |
|          | b) An array stores elements in _____ locations.                                            | 1     | 3  | 2  |       |
|          | c) A structure is defined using the keyword _____.                                         | 1     | 3  | 2  |       |
|          | d) Pointer variable stores _____.                                                          | 1     | 3  | 2  |       |
|          | e) The operator used for pointer dereferencing is _____.                                   | 1     | 3  | 2  |       |
|          | f) The statement <code>if(a == b)</code> compares _____ of a and b.                        | 1     | 3  | 2  |       |
|          | g) In <code>for(i=0; i&lt;5; i++)</code> , the loop executes _____ times.                  | 1     | 3  | 2  |       |

OR

- (e) a) What will be the output of the following C code? Explain with proper justification. 3 3 6 1.4.1

```
#include (stdio.h)
int main()
{
    int i, j;
    for (i = 2; i < 10; i++)
    {
        for (j = 2; j <= 5; j++)
        {
            if (!(i % j))
                break;
            if (j <= i / j)
                printf("%d ", i);
        }
    }
    return 0; }
```

- b) Explain call by value and call by reference with an example of code. 4 3 4 1.4.1
- Q4 (a) Define primary key and give one example. 2 4 2 1.4.1
- (b) Differentiate between DDL and DML. 2 4 2 1.4.1
- (c) Compare file system vs DBMS. 3 4 5 1.4.1
- (d) Fill in the Blanks 1.4.1
- a) DBMS stands for \_\_\_\_\_. 1 4 2
- b) ER stands for \_\_\_\_\_. 1 4 2
- c) DBA stands for \_\_\_\_\_. 1 4 2
- d) To extract and display table information we use the \_\_\_\_\_ command. 1 4 2
- e) A primary key uniquely identifies each \_\_\_\_\_. 1 4 2
- f) A foreign key refers to the primary key of another \_\_\_\_\_. 1 4 2
- g) A set of entities of same type forms a \_\_\_\_\_. 1 4 2

OR

- (e) Write and explain INSERT, UPDATE, DELETE with an example of each. 7 4 4 1.4.1
- Q5 (a) Define Artificial Intelligence. 2 5 1 1.4.1
- (b) Explain supervised learning with an example. 2 5 2 1.4.1
- (c) Describe neural networks with suitable examples. 3 5 3 1.4.1
- (d) a) Explain applications of AI in healthcare and cybersecurity. 4 5 4 1.4.1
- b) Evaluate the future impact of AI on jobs. 3 5 5 1.4.1

OR

- (e) Fill in the Blanks 7 5 2 1.4.1
- a) Machine learning allows computers to learn from \_\_\_\_\_. 7 5 2 1.4.1
- b) A neural network is inspired by the human \_\_\_\_\_. 7 5 2 1.4.1
- c) Deep learning uses multiple \_\_\_\_\_ layers. 7 5 2 1.4.1
- d) Labeled data is required in \_\_\_\_\_ learning. 7 5 2 1.4.1
- e) Unlabeled data is used in \_\_\_\_\_ learning. 7 5 2 1.4.1
- f) A machine that performs human tasks is called an \_\_\_\_\_ system. 7 5 2 1.4.1
- g) Decision-making based on rewards is called \_\_\_\_\_ learning. 7 5 2 1.4.1

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